

B. Addition on a Segment

time limit per test: 2 seconds

memory limit per test: 512 megabytes

You start with an integer array a , which initially consists of n zeros. You have to perform the following action **exactly** n times:

- choose two integers l and r ($1 \leq l \leq r \leq n$) and assign $a_i = a_i + 1$ for each i such that $l \leq i \leq r$.

You are given an array b , consisting of n integers. Your task is to choose such values l and r for each action that:

- after all n actions are performed, it's possible to reorder the elements in such a way that a becomes equal to b ;
- the maximum value of $r - l + 1$ over all actions is as large as possible.

What's the maximum possible value of $r - l + 1$?

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains one integer n ($1 \leq n \leq 2 \cdot 10^5$) — the length of the array b .

The second line of each test case contains n integers b_i ($0 \leq b_i \leq n$) — the elements of the array b .

Additional constraints on the input:

- there exists at least one way to choose l and r for each action and reorder the elements at the end so that a becomes equal to b ;
- the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output one integer — the answer to the problem.

Example

input	Copy
3 5 0 5 1 0 1 3 3 2 1 5 1 1 1 1 1	
output	Copy
3 3 1	

Note

Consider the first test case. If the n actions were as follows:

- $l = 3$ and $r = 3$
- $l = 1$ and $r = 3$
- $l = 3$ and $r = 3$
- $l = 3$ and $r = 3$
- $l = 3$ and $r = 3$

The array $a = [1, 1, 5, 0, 0]$, so you can reorder the elements to make it equal to $[0, 5, 1, 0, 1]$. As can be seen in this case, the maximum value of $r - l + 1$ is 3. It can be shown that this is the

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Language: GNU G++17 7.3.0

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optimal answer.

In the second test case:

- $l = 1$ and $r = 3$
- $l = 2$ and $r = 3$
- $l = 3$ and $r = 3$

The answer is 3.

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